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Performance and Proximity Investigations on Small Scale Lensed Turbines

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Abstract

Aerodynamic interactions can lead to increase in power generated from a wind lens turbine by 20%. This work investigates the optimal design of a small scale, low Reynolds number, 1-foot diameter wind lens turbine (WLT) and the aerodynamic interactions between wind lens turbines in all three axes. Multiple rotor designs and lens-lens interactions were experimentally investigated at the University of Dayton Low Speed Wind Tunnel (UD-LSWT). A full scale grid array of 12 WLTs was designed, manufactured, and field tested at the University of Dayton. The iterative blade design process yielded a WLT with an efficiency of 10.6% based on measured electrical power. Force-based testing on circular disks (surrogate to lensed turbines) and electric power testing on WLTs was conducted in a 1-D (side-by-side) and 3-D (upstream-to-downstream) grid configurations to gain insight into lens-lens aerodynamic interactions. Reasonable correlations were observed between the disk and the WLT testing in 1-D configuration. The WLTs yielded a 16% increase in coefficient of power at 0.1 X/D. The 3-D WLT results indicated an inherent bias in the downstream WLT power production towards the retreating blade side of the lead WLT. The results from field testing of the full scale grid array exhibits promise as a cost-effective alternative to large scale wind turbines.

I. Nomenclature

A	=	Reference Area (m^2)
$\overline{C_D}$	=	Time Averaged Coefficient of Drag
C_P	=	Coefficient of Power
$\overline{D}$	=	Time Averaged Drag Force (N)
P	=	Electric Power (W)
ρ	=	Density (kg/m^3)
ω	=	Rotor RPM (rad/s)
R	=	Rotor Radius (m)
U_∞	=	Freestream velocity (m/s)
TSR	=	Tip Speed Ratio $(\omega R/U_\infty)$

II. Introduction

One of the greatest challenges of the 21st is the development of a low carbon energy source. While oil, coal, and gas have been critical to meeting the energy demands of the growing population, it has led to the degradation of

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the air quality and is causing irreparable damage to the climate. As a result, there has been a push for development of clean and renewable energy source including wind, solar, hydroelectric, geothermal, and nuclear. Wind is expected to supply up to 35% of the nation's energy requirements by 2050 [1]. While wind turbines have been used for generations, there are factors which hinders the adaptation of the turbines by the general public. In order to produce more power, wind turbines have been consistently increasing in size. Current wind turbines have rotor diameters in excess of 150 meters [2] which makes them larger than wings of some commercial airliners. Because of the sheer size, wind turbine implementation is a capital-intensive process due to the high manufacturing, transportation, and installation costs.

An alternative solution to the use of large-scale wind turbines is the wind lens turbine (WLT). A wind lens turbine is similar to a traditional wind turbine except the blades are surrounded by a shroud of a particular shape. This shrouded device accelerates the air passing through the rotor and has been shown to increase the turbine power output by a factor of 2 to 5. A cross-sectional view of a wind lens turbine (WLT) is displayed in Fig. 1. The brim of the lens creates a recirculation region behind it where low pressure is generated. The low pressure accelerates the flow through the rotor which creates a higher effective freestream (wind) velocity.

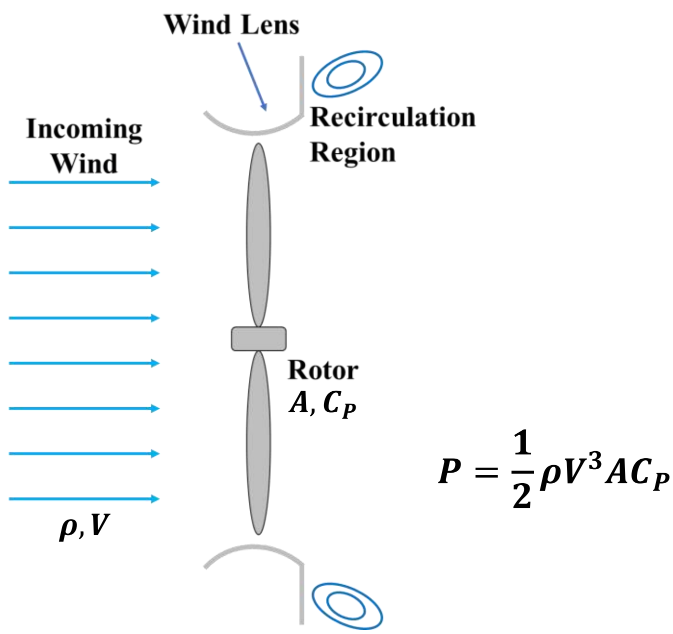


Fig. 1 Schematic of Wind Lens Turbine

Power generated by a wind turbine is proportional to the freestream velocity cubed. Therefore, if the wind speed doubles, the power generated increases by a factor of eight. Because the wind lens turbine (WLTs) can produce 2 to 5 times more power, they can be made 2 to 5 times smaller and still maintain equivalent power output as a traditional wind turbine. Therefore, the current research explores a novel idea of combining multiple lensed turbines in a grid structure to harness wind energy for residential, commercial and military applications. However, when multiple lenses are placed in a grid, the effectiveness of the lens, and hence the power output can be increased when compared to an individual lensed turbine.

2.1 Wind Lens Turbine Viability:

Extensive investigations on the lensed turbines was done by Ohya et al. [3], [4], [5], [6]. Ohya and Karasudani tested two types of wind lenses: a long type diffuser and compact style diffuser. The experimental work displayed a 4 - 5 times power increase in the long type lens while the compact style lens displayed a 2 - 3 times increase in power. These increases were attributed to the formation of vortices behind the brim. The vortices generate an area of low pressure drawing more air through the rotor. This work showed that the lensed turbine can produce much more power than a traditional style wind turbine. The conclusion was drawn that the long style diffuser produces more power than the compact style diffuser, however, the compact style diffuser is far more practical.

2.2 Selection of a Wind Lens Profile:

Khamlaj and Rumpfkeil [7] established a high-fidelity CFD framework to develop a high power, low drag wind lens which was optimized for a wind speed of 8 m/s. The optimized lens profile was validated by comparing to the previously mentioned work done by Ohya and Karasudani [3]. The results of this work displayed that wind lens turbines could be improved upon through the use of this computational model. The results of the work yielded three lens profiles that had a higher coefficient of power than previously developed models. The profile selected for the work in this paper was the lens profile that maximized the coefficient of power and minimized coefficient of drag. This profile is displayed in Fig. 3 which compares the optimized lens to the conventional lens used in testing. Performing hot-wire anemometry testing at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) behind the conventional lens and optimized lens, the optimized lens revealed a higher Strouhal number with respect to the Reynolds number. This variation is shown in Fig. 4. This increased frequency of vortex shedding allows for the optimized lens to perform better than the conventional lens investigated in [3].

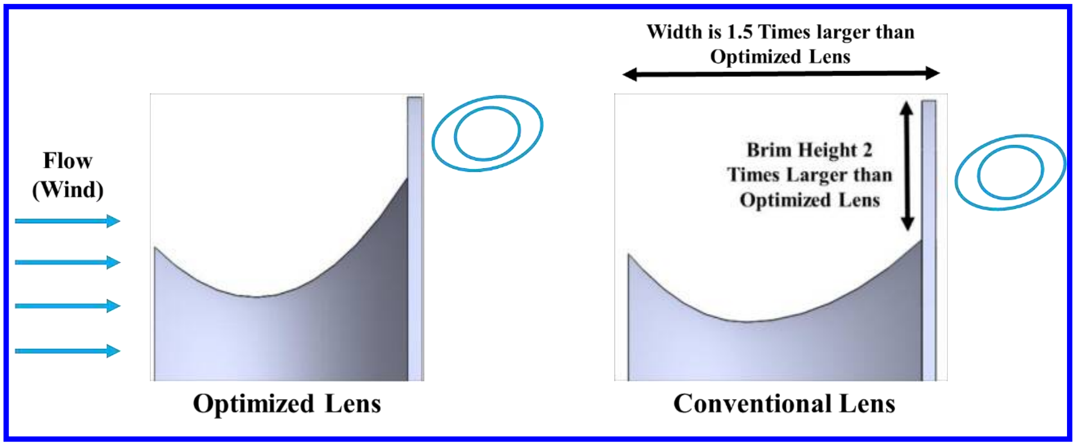


Fig. 2 Optimized (from *Khamlaj and Rumpfkeil [7]*) and Conventional (from *Ohya and Karasudani [3]*) Lens Profile

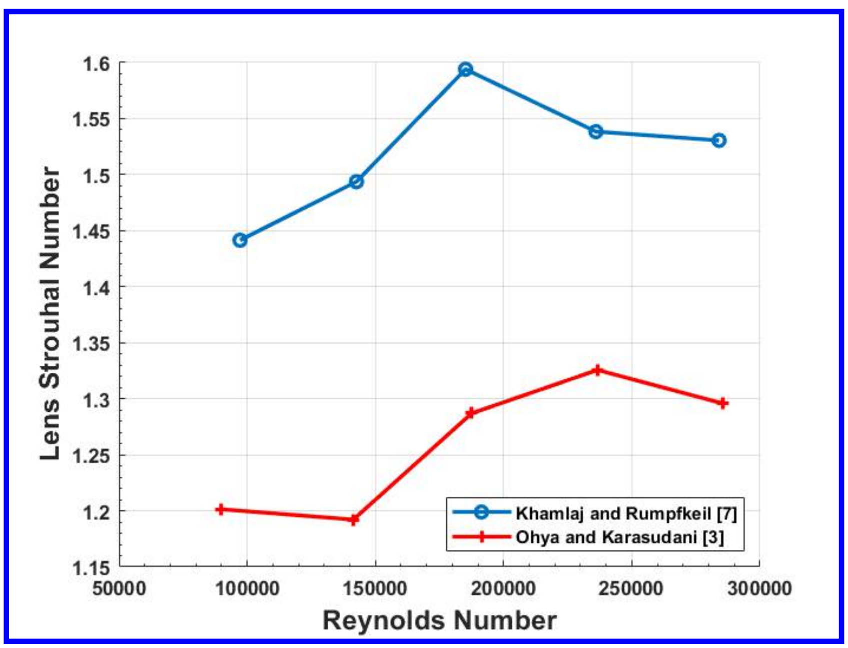


Fig. 3 Variation of Lens Strouhal number vs Reynolds number for the optimized *(from* Khamlaj and Rumpfkeil [7]) and the conventional lens *(from* Ohya and Karasudani [3]).

2.3 Examination of Multirotor Work:

A multirotor wind turbine is a wind turbine that is made up of multiple rotors in place of one larger rotor. Jamieson [8] presented the idea of a multi rotor wind turbine. In this examination, he stated that multirotor technology has a long history and persists in a variety of modern innovative systems but has fallen out of consideration in mainstream design due to its perceived complexity. Jamieson emphasizes that the central advantages of multirotor systems is the ability for standardization of parts. Standardization of wind turbine parts allows cost to be driven down while increasing quality. In addition, larger capacities could be obtained with more rotors instead of larger rotors. Also, multi rotor systems can reduce uneven loads on single large turbine blades and reduce the net torque the support structure experiences. Also, failure of a rotor results in a 1/n reduction in capacity instead of total failure in a traditional turbine. Weight is reduced by $1/\sqrt{n}$ where n is the number of rotors in a multi-rotor system

Habib et al. [9] reviewed the advantages and disadvantages of a multi rotor wind turbine system which revealed that a 4-rotor x 5 MW and a 45-rotor x 444 KW reduced cost by 20% and 30% respectively, when compared to a baseline 20 MW traditional turbine. A work performed by the Southwest Research Institute [10] showed that a full-scale test of a seven-rotor system at the NASA Langley wind tunnel had no negative effect on power produced. Further, the rotors showed a 2-16% increase in power depending on spacing. Thus, it has been shown that multirotor systems can generate equivalent or better power while lowering cost through standardization of parts. Therefore, similar inference can be made for multiple lens turbines in a grid.

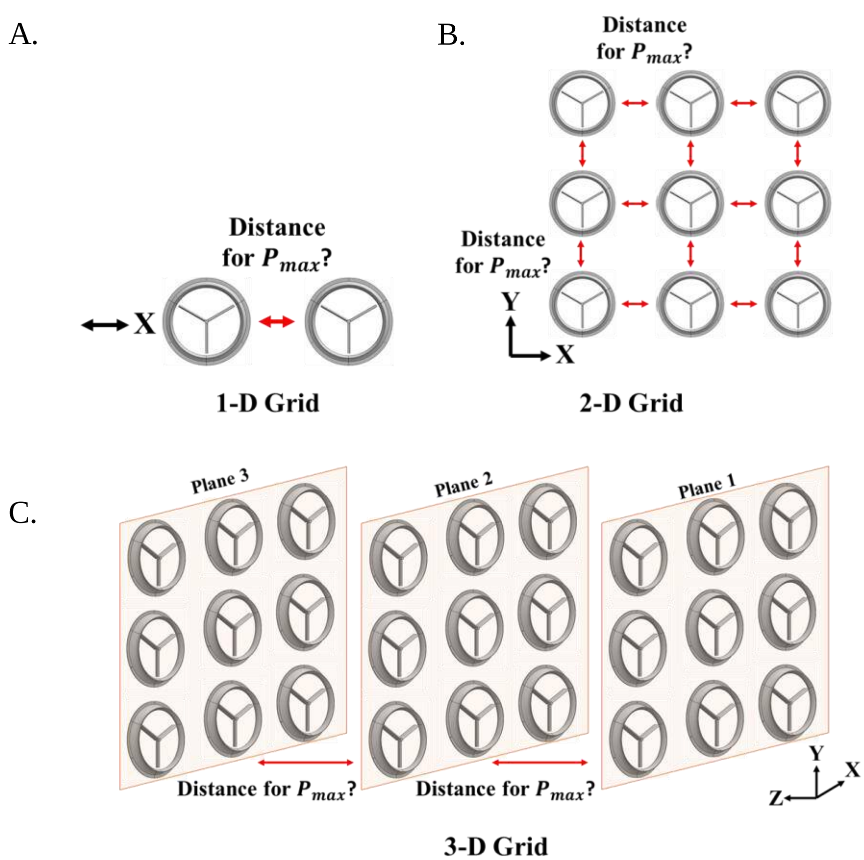


Fig. 4 Schematic of a) 1-D grid b) 2-D grid and c) 3-D grid

Göltenbott et al.et al. [11] and Ohya et al. [4] examined the spacing of wind lens turbines and the effect on aerodynamic performance. The results of the experiment noted that an increase in power was seen from the side-by-side multirotor system (1-D configuration). The power increase was dependent upon the gap ratio: the measure of the distance between the wind lenses to the diameter of the wind lenses. Göltenbott discovered that two wind lenses placed side-by-side produced a power increase of 3-5% while three wind lenses placed side-by-side produced a net power increase of 5-9%. Ohya also showed that a 12% increase in net power can be obtained in three lens setups placed side-by-side. Watanabe and Ohya [5] also experimented with a 2-D grid consisting of five WLTs. The five-rotor system contained three rotors in a side-by-side arrangement with a two-rotor side-by-side arrangement above. This work revealed a total power increase of 21% for the five-rotor setup. Similar investigation is shown in this paper for 12 turbines in 2-D configuration.

The objective of the research presented is to test the performance of the small scale WLTs individually and at close proximity to other lenses in 1-D, 2-D and 3-D grid format as shown in Fig. 4, ultimately leading to a system that is lower cost, easier to transport, easier to manufacture, and easier to install. A 1-D grid consists of two or more WLTs in close proximity in one axis. This can be side-by-side as shown in the X-axis or top-and-bottom in the Y-axis. A 2-D grid consists of several WLTs in the X-axis and Y-axis with turbines in a side-by-side and top-and-bottom configuration. The 3-D grid consists of WLTs in all in the X, Y, and Z axes. This style of grid system makes use of multiple 2-D grids in successive planes. The benefit of utilizing a 3-D grid system the ability to extract more energy from a given land space. Although downstream turbines produce less power, the configuration increases the net power output in a significantly smaller foot-print.

The following experimental objectives were conceived and implemented to satisfy the research objectives:

1. Optimize rotor geometry for maximum electrical power output for small scale (1ft in diameter) lensed turbine using Q-blade software [13] and experimental testing with variable parameters such as airfoil geometry, blade incidence angle, root twist, tip twist, number of blades, taper/inverse taper ratio.
2. Characterize aerodynamic interaction by quantifying changes in coefficient of drag of individual circular disks (acting as a surrogate for lensed turbines) in 1-D, 2-D and 3-D grid configuration.
3. Replace disks with WLTs in 1-D, 2-D and 3-D configuration and compare the individual and net power output with results from disk testing
4. Field test WLTs in a 1D, 2D, and 3D grid system at the University of Dayton

III. Optimal Rotor Design

A first order approximation was conducted prior to the design of the rotor for testing. The expected blade tip Reynolds Number was approximately 14,000 whereas full-scale wind turbines have Reynolds numbers in excess of 1,000,000. This was estimated at a freestream velocity of 8 m/s and a rotational speed of 300 RPM (approximately 0.5 TSR). Once this was known, a suitable airfoil was selected. To determine the best airfoil for the turbine blade at low Reynolds number, a viscous XFOIL sensitivity was conducted on several airfoils including NACA and S-series to determine the highest lift to drag (L/D) ratio as shown in Fig. 5. Airfoils of varying thickness and camber were examined. It was found that a thin and highly cambered airfoil was the best choice for the turbine. The NACA 6604 and NACA 2404 were found to have the highest peak aerodynamic efficiency. However, the NACA 6604 is more suitable because it has a smoother peak. Additionally, it has been suggested by Manwell et al. [12] that for TSRs below 3 that a curved plate can be used in place of an airfoil. Therefore, a curved plate was selected for the experimental work as it closely resembles the NACA 6604. Further, for additive manufacturing purposes, it is more suitable for printing.

The software package Q-blade [13] was used in the design of the turbine rotor. Q-blade is able determine the coefficient of power versus tip-speed ratio for a given airfoil profile, angle of attack, number of blades, and other parameters such was blade twist and taper ratio. Several design iterations were developed using Q-blade and tested experimentally in the UD-LSWT as shown in Table 1. The rotor design was conducted in Q-blade for a specific TSR as shown in the table. The rotor was then designed, manufactured, and tested in the UD-LSWT. As the performance of the rotor changed, the design TSR was modified and the process was repeated. On iteration three, the design TSR and experimental TSR converged to four and the process was completed. The final design of the optimized rotor is shown in Fig. 6.

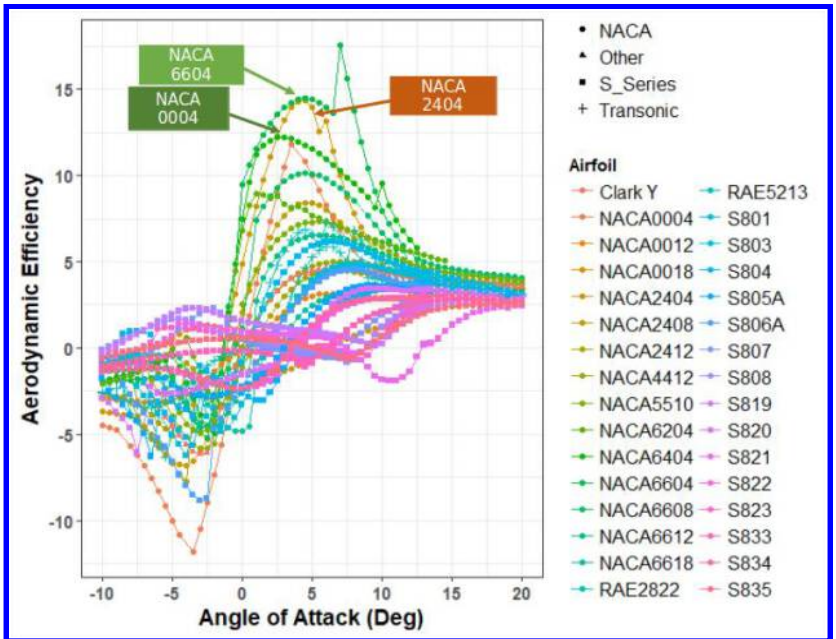
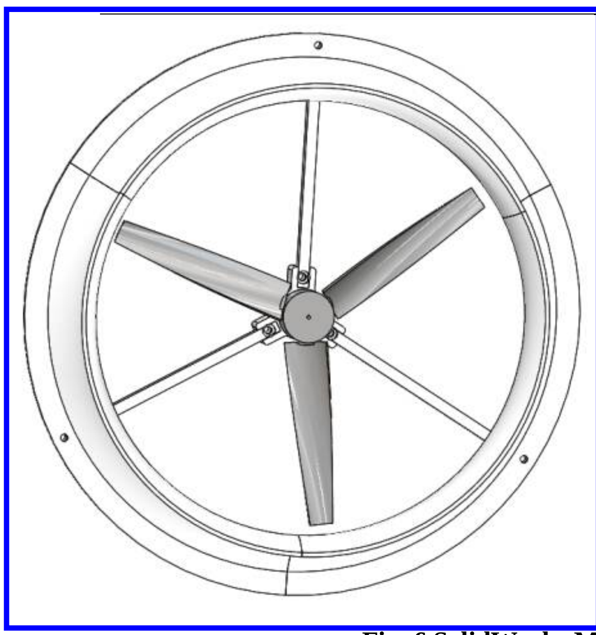


Fig. 5 Aerodynamic Efficiency of Various Airfoils at a Reynolds Number of 14,000

Table 1 Rotor Properties of various Design Iterations

Iteration Number	Design TSR	Root Pitch	Tip Pitch	Blade	Number of Blades	Experimental TSR
1	0.5	45°	45°	Curved Plate	3, 5, 7	1.8
2	2.0	45°	6.93°	Curved Plate	3, 5, 7	3.5
3	4.0	45°	-1.93°	Curved Plate	3	4.0



Blade: Curved Plate with 5% camber at 50% chord

Chord: 1 inch

Taper Ratio: 0.5

Root Pitch: 45°

Tip Pitch: -1.93°

Fig. 6 SolidWorks Model of Final Rotor Design

IV. Experimental Setup

A. Wind Tunnel

All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The UD-LSWT has a contraction ratio of 16:1 with 6 anti-turbulence screens. The freestream turbulence intensity is below 0.1% at 15 m/s measured by hotwire anemometer. The test section velocity varies from 3 m/s to 40 m/s with four interchangeable test sections. The tunnel also contains a downstream shuttering system that varies the freestream velocity profile up to 5 Hz and 50% of the velocity amplitude.

All the experiments mentioned in the paper were done in the open jet configuration. The test section inlet measures 76.2cm x 76.2cm (30in x 30 in) and opens into a pressure sealed plenum. The effective test section length is 182 cm (72 in). A 112cm x 112cm (44 in x 44 in) collector returns the expanding air to the diffuser. The freestream velocity is measured using a Pitot tube connected to a TSI Micromanometer (Model 5825).

B. Rotor Testing

The rotor testing in the University of Dayton Low Speed Wind Tunnel (UD-LSWT) was necessary to converge to an optimal rotor design shown in Fig. 6. In each iteration shown in Table 1, the rotors were tested independently in the test section of the UD-LSWT as shown in Fig. 7.

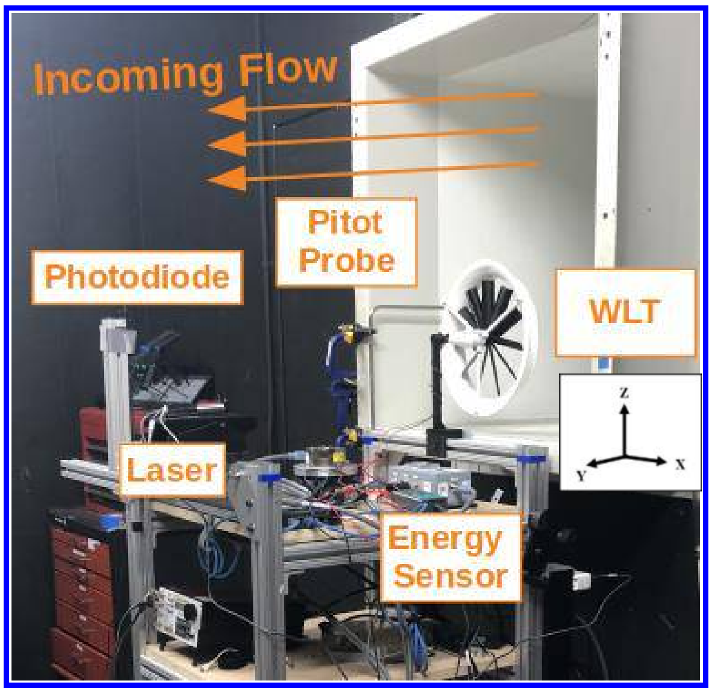


Fig. 7 Rotor Testing in the University of Dayton Low Speed Wind Tunnel

The performance of rotors with different number of blades was evaluated by examining the coefficient of power and tip-speed ratio. The initial parameters changed for testing was the number of blades and the external resistance. Because of the size of the rotors in this work are comparatively smaller, the tip-speed ratio and the blade Reynolds number were expected to be low because TSR is a function of the turbine radius. Therefore, experimental work was conducted to determine the relationship between the number of blades and power production. The test matrix for the experimental work is displayed in Table 2. As the electrical power was being measured and not mechanical power, the power production is dependent on a couple parameters: 1) The rotational speed the generator is operating at. 2) The external resistance the power is being measured across. It was assumed that the changes in generator efficiency for different rotational speeds were marginal. The external load will also change the power generated by the generator. A sensitivity of three different resistances was conducted to see how power production varied with the external load. It was found during testing that a 1 Ohm resistor yielded peak power results. Each rotor was tested in a freestream condition of 0-10 meters per second to examine the power versus freestream and rotational speed versus freestream. The power output was gathered at randomized freestream velocities to avoid bias and creep. The velocity was varied in 0.5 meter per second increments across the freestream range and each point was repeated twice for a total of 20 test points per rotor.

Table 2 Rotor Test Matrix

Number of Blades	Freestream Velocity [m/s]	External Resistance [ohms]	Power Sampling Rate [Hz]	Power Sampling Time [s]
3, 5, 7	0-10	1, 10, 30	250	30

C. 1-D and 3-D Circular Disk Proximity Testing

Disk testing was conducted to explore the aerodynamic interactions between the turbine lenses at close proximity to one another. The disks act as a surrogate to lenses experiencing similar aerodynamic effects which can be quantified by changes in coefficient of drag. Two disks were used in testing and the baseline coefficient of drag was benchmarked prior to the proximity testing. The percent change in coefficient of drag for Disk 1 and Disk 2 was calculated using Equation 1.

$$\overline{\%\Delta C_{D1,2}} = \frac{\overline{C_{D_{1,2}}} - \overline{C_{D_\infty}}}{\overline{C_{D_\infty}}} * 100 \tag{1}$$

A positive change in the coefficient of drag represents an increase in drag when compared to a standalone test, whereas a negative change in coefficient of drag represents a decrease in drag on the disk. An increase in drag suggests potential for higher energy extraction from the turbine, assuming no changes in local dynamic pressure and consequently, provides better flow for the turbine.

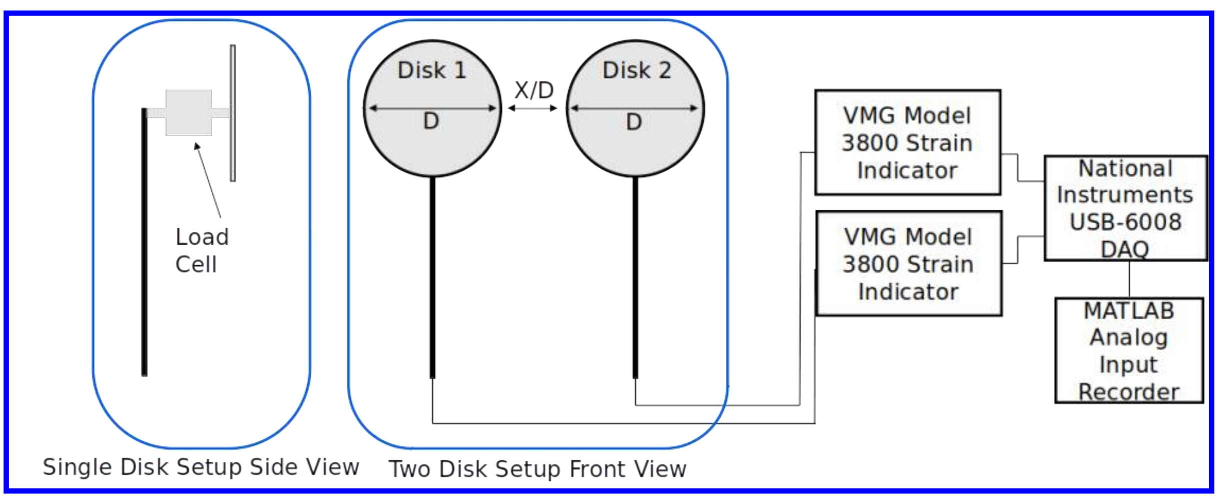


Fig. 8 Baseline (left) and 1D Disk (Right) Testing Setup along with Equipment Used.

Disk testing was conducted in three primary steps: baseline testing, 1D, and 3D proximity testing. Baseline testing was conducted on each of the two disks used. The drag was measured individually using the Honeywell Model 31 (+/- 10 lbs, 0.25% accuracy at full-scale) threaded load cell at 1000 Hz for 120 seconds to ensure good baseline data. Once the baseline data was established, the 1D proximity testing was conducted. Both disks were placed into the flow at varying gap ratios (X/D) which is a non-dimensionalized spacing of the disks. This distance is defined as the distance between the disks divided by the diameter of the disks. The disks were varied in terms of spacing from 0.0 X/D (touching) to 1.5 X/D (1.5 diameters of space between the disks) in increments of 0.1 X/D. The drag on each disk was measured at 1000 Hz for 120 seconds. A schematic of the test setup for baseline testing and 1D testing is displayed in Fig. 8.

The 3D disk testing included varying the distances in the stream-wise direction as well as the cross-stream direction. In this test setup, the disks were varied from 0.5 X/D to 8.0 Y/D in increments of 0.5 Y/D in the downstream direction. Then, the disks were offset in the cross-stream direction by increments of 0.5 X/D from -2.0 X/D to 2.0 X/D. Data was collected at 1000 Hz for 30 seconds in this case. A schematic of the test setup is displayed in Fig. 9. The freestream velocity for the 1D and 3D cases was 24.00 m/s and 22.42 m/s, respectively, yielding Reynolds numbers based on the diameter of the disk of 160,000 and 150,000. These Reynolds numbers are similar to the Reynolds number experienced by the lensed turbines based on diameter (1 ft. diameter turbine at 6 m/s freestream is 150,000).

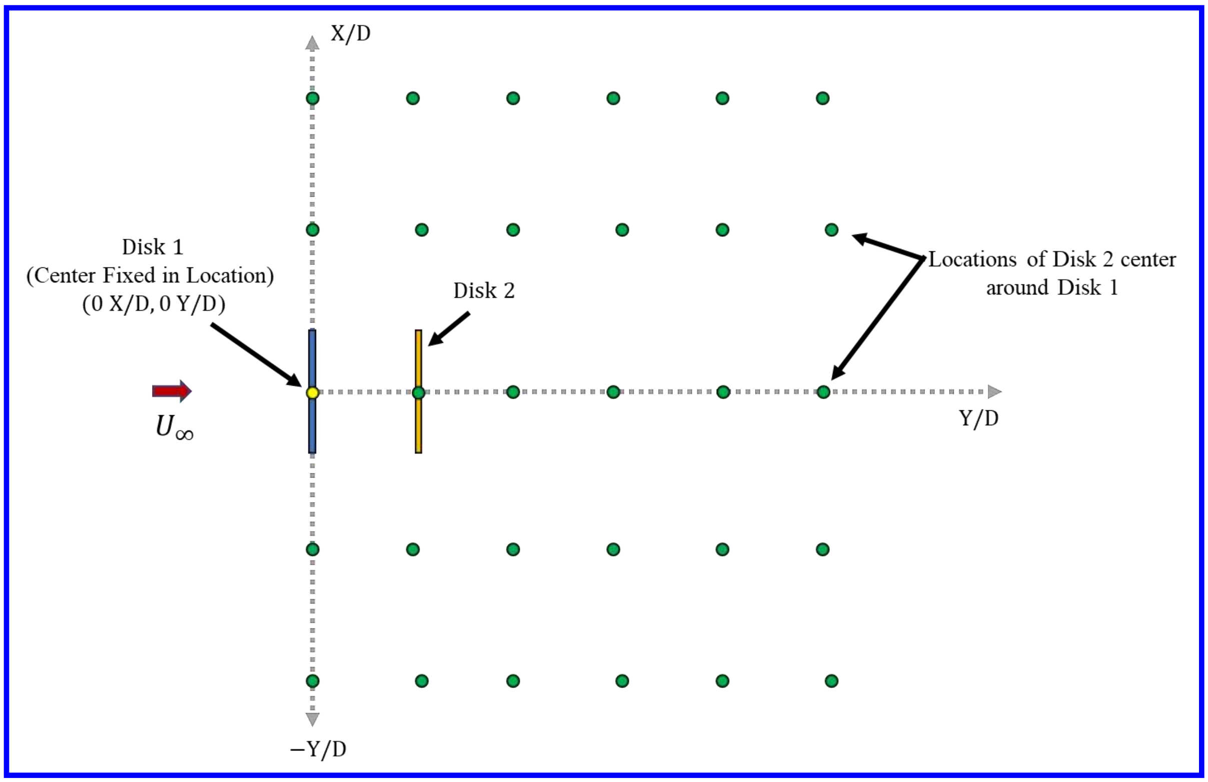


Fig. 9 Schematic showing the 3D Disk Testing Setup. Similar setup was used for 3D WLT Test.

D. 1-D and 3-D WLT Testing

The testing method for the 1-D and 3-D rotor testing is quite similar to the testing method for the disks. However, instead of disks, WLTs are used. The freestream velocity in both these cases was 7.5 m/s. An image of the WLTs in proximity is shown by Fig. 10. The non-dimensionalized spacing X/D is also used in this case where X is the space between the turbines and D is the diameter of the turbines. Prior to testing the WLTs in proximity, each WLT was tested individually at different test section locations to quantify the influence of wall effects. Once the turbines baseline performance was quantified the proximity tests were conducted. The turbines were spaced from X/D of 0.0 to 0.5 in increments of 0.1 X/D. Testing was also done with the location of the turbines interchanged to remove any inherent bias. The tests were conducted in two parts. The test was conducted once with the turbines tested at each X/D value. Once the test was finished, the two turbines swapped positions. The results at each position were then averaged to determine the coefficient of power versus X/D spacing for the left and right turbine. The 3D WLT testing was conducted at spacings of -0.5 to 0.5 X/D and 0.5 to 2.0 Y/D in increments of 0.5 X/D or Y/D in a structure similar to the disk testing as shown in Fig. 9.

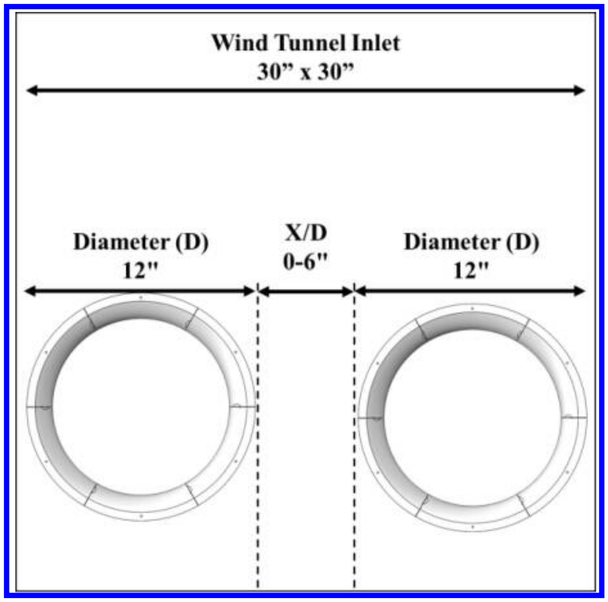


Fig. 10 WLT 1D Testing

E. Field Testing

A 2D grid array of 12 WLTs was constructed at the University of Dayton as shown in Fig. 11. The structure is setup in a 3x4 fashion (3 rows, 4 columns). In the structure, the WLTs are spaced at a non-dimensional spacing of 0.25 x/D, or 3 inches. The top of the structure contains equipment for monitoring and measuring the local wind speed. A Calt 3 cup anemometer was used to measure the oncoming wind speed and a SwitchDoc Labs Wind Vane was used to measure the direction of the wind. The structure is mounted on a thrust bearing allowing the grid to yaw towards the direction of the wind. The structure is capable of rotating +/- 30° from the its neutral facing direction. The waterproof base contains data logging equipment for capturing the wind speed, wind direction, and power output of the grid. The data was logged using an Arduino Uno and the current was measured using a NOYITO current sensing module. Alternatively, the data logging equipment in the base can be replaced with a standard 12-volt battery for energy storage.

Each arm on the grid structure connects its turbines electrically in series and each arm on the structure is connected in parallel. The arms are connected in series to boost the voltage output and the arms are connected in parallel to provide robustness to the structure. If a turbine is damaged and knocked offline on an arm, the remaining arms can still generate power. The electrical load passes through a series of resistors totaling 2 ohms and then the current is measured and recorded.

F. Test Models

All components required to satisfy the experimental objectives were modeled using SolidWorks. The turbine consisted of a Vernier High Torque generator (KW-HIGEN), a blade hub, turbine blades, and a central mounting with support. The rotor diameter for the turbine was selected to be 22.9cm (9 in). For testing of the wind-lens, the profile developed by Khamlaj and Rumpfkeil [7] was modeled and arm supports were used to mount the lens to the central hub. The diameter of the wind lens was selected to be 12 in (30.5cm). The disks were modeled with a diameter of 4.28 inches (10.87cm) and a depth to diameter ratio (d/D) of 5%. All models were printed using a GorillaMaker GM3D500 printer and were sliced using Simplify3D software. PLA plastic was used for each component. Any parts that had a poor surface finish were carefully sanded using a series of sandpapers (200, 400, and 800 grit) to remove surface defects.

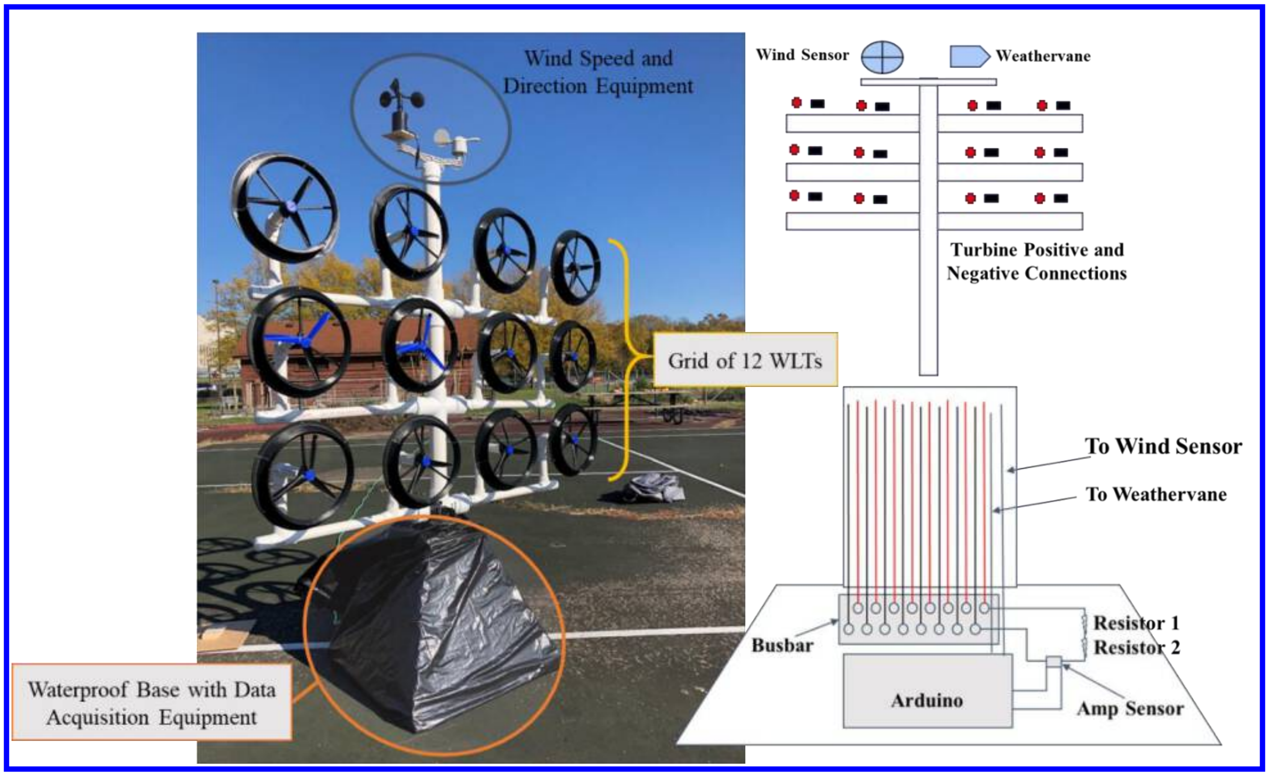


Fig. 11 Experimental Setup for Grid Array of 12 WLTs Located at the University of Dayton Organized in a 3x4 Orientation.

G. Electrical Generator and Force-Based Testing

To measure the power production from each turbine, an electrical generator was used. Henceforth, any mention of power production represents the electrical power. The electrical generator selected for this experimental work was the Vernier High Torque generator (KW-HIGEN). This generator was selected because it is tailored for small scale wind experiments. The generator has a rated voltage of 12.0 volts, rated load of 45 g-cm, a maximum power output of 11 watts, and a measured internal resistance of 1 ohm. The generator is connected to a Vernier Go Direct Energy Sensor (GDX-NRG) with a built-in internal resistance of 30 ohms. The energy sensor (resolution of ± 1 mV for voltage and ± 40 μA for current) has the option to attach an external load to the sensor. The max external voltage is 30V and current is 1A.

The drag on the disks was measured using an analog Honeywell Model 31 (+/- 10 lbs, 0.25% accuracy at full-scale) threaded load cell. The sensor is a uniaxial load cell capable of measuring tension and compression. The load cell was wired to a Vishay Measurements Group Model 1800 signal conditioner/amplifier using a gage factor range of 0.24-0.50 and excitation voltage of 5V. The signal was then connected to a National Instruments USB-6008 DAQ. The NI DAQ has 12-bit resolution with a max sampling rate of 10 kHz. The signal is then recorded using the MATLAB Analog Input Recorder and post-processed. The two sensors were then calibrated using 20 different loading points.

V. Results

A. Rotor Testing and Sensitivity on Blade Taper Ratio

The experimental work conducted at the UD-LSWT on a single lensed turbine is presented in this section. Fig. 12 shows the rotational speed versus wind speed, electrical power versus wind speed, and coefficient of power versus tip speed ratio for the optimized rotor-lens configuration shown in Fig 6. The results of the rotational speed show that the tapered blade has an increased rotational speed compared to the straight blade and inverse taper. At a freestream velocity of 5 m/s, the rotational speed of the turbine is 2000 RPM. This is high for a wind turbine as most large scale wind turbines operate at 30-60 RPM. One practical benefit of the increased rotational speed is the increased operating efficiency of the generator negating the need for a complex gear train. However, there is a tradeoff to be made in terms of centrifugal forces on the blades, safety, and wear on the system. With respect to the electrical power results, the tapered blade displays the best performance beyond 4 m/s. Below 4 m/s, the tapered blade and straight blade exhibit comparable performance, both of which exceed the inverse taper blade.

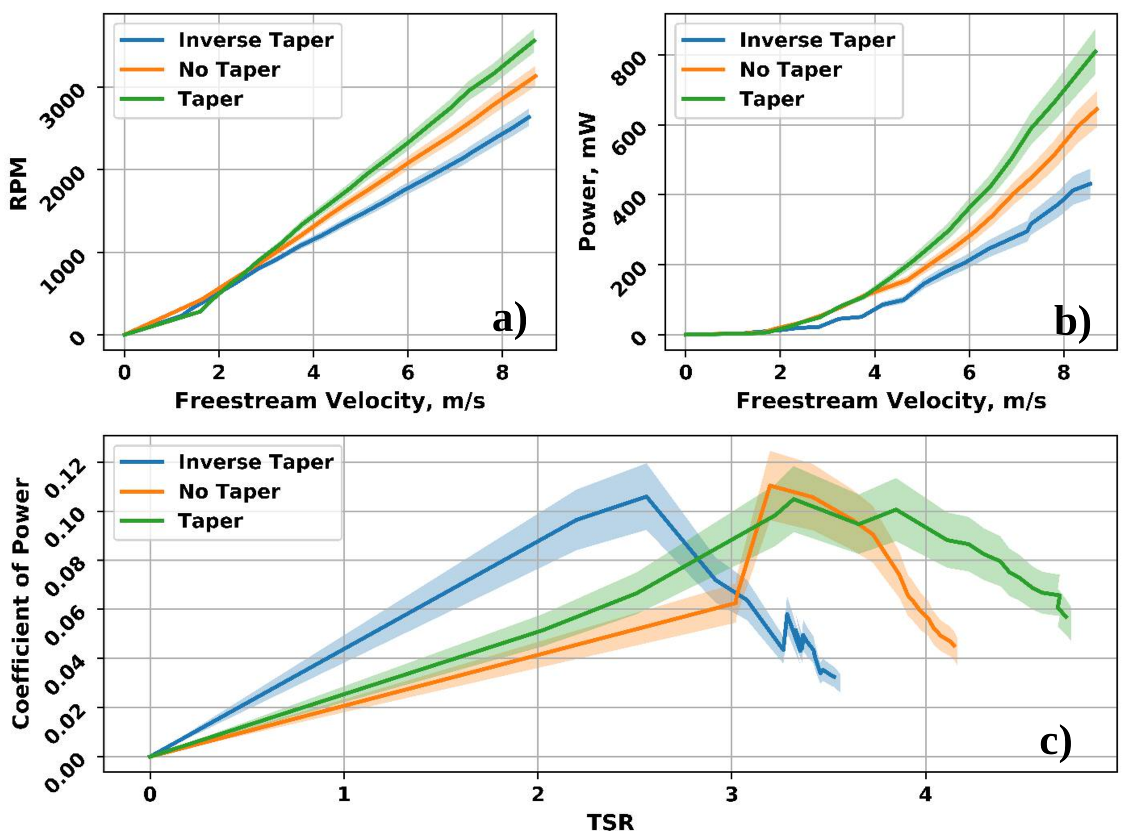


Fig. 12 a) Rotational Speed of WLT b) Power of WLT c) Coefficient of Power versus TSR for WLT

As the wind speed and rotational speed increase, the tapered blade becomes more effective. This is likely due to the decrease in the blade tip vortex formation that is associated with a smaller tip chord. The non-dimensionalized results of the experiment are displayed in Fig. 12c which shows the coefficient of power versus tip-speed ratio. It should be noted that the rotors were allowed to rotate freely and were not controlled by any device other than the generator. Further, because of the simple design, there is no active pitch control system. This means that the tip-speed ratio could not be controlled and is simply a function of the rotor design. The peak coefficient of power of 11.0% is observed with the straight blade rotor. The peak coefficient of power for the inverse taper blade and tapered blade is 10.6% and 10.5%, respectively. Although the straight blade produces the highest coefficient of power, for practical applications and use in a grid structure, the tapered blade provides the best performance because it displays a more gradual reduction in coefficient of power with increase in tip speed ratio. Therefore, as the wind speeds change, the tapered blade is more adaptable to changing wind speeds. This result is further emphasized in the power results. At low wind speeds, the tapered blade and straight blade display comparable performance, but as the wind speed increases, the tapered blade outperforms the straight blade. This result led to the selection of the tapered blade for use in the rotor.

B. 1-D Testing

a) 1-D Disk Testing (Side-to-Side Arrangement of Disks)

To evaluate the flow characteristics of bluff bodies in close proximity, disk testing was conducted in 1D and in 3D as shown in Fig. 9. The results of the 1D test are displayed in Fig. 13 and includes comparison data of work done by Ohya et al. [4]. The percent change in coefficient of drag from baseline is displayed for the two disks, as well as the average percent change in coefficient of drag shown in Equation 1. The results display reasonable agreement with the work of Ohya et al. [4], particularly from X/D 0.0-0.2. At the range of X/D 0.3-0.5, there is a slight magnitude discrepancy, but the overall trend is the same. An interesting flow feature is the oscillatory nature in the percent change in drag of the two disks at low gap ratios (X/D < 0.6). This suggests that there is an inherent bias to the flow at close proximity. This phenomenon was also documented in Ohya et al. [6]. The net increase in drag on the lens grid system suggests that WLTs in close proximity can see an increase in power even at smaller scales.

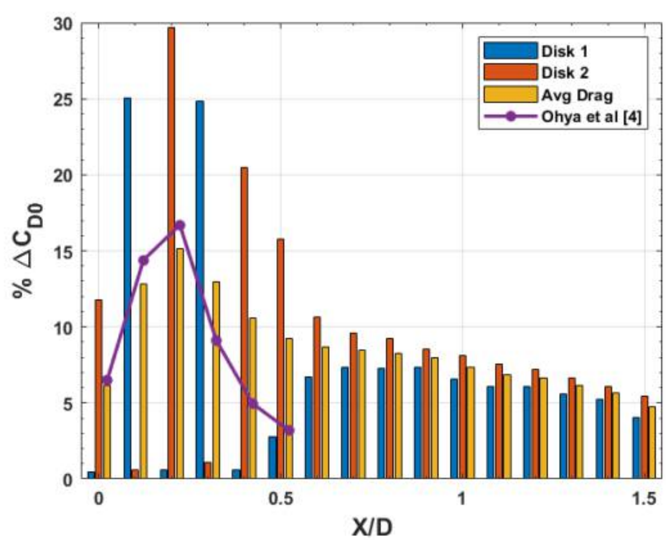


Fig. 13 Percent Change in Coefficient of Drag for 3D Disk Testing

b). 1-D WLT Testing (Side-to-Side Arrangement of WLT)

The result of the 1D WLT testing is displayed in Fig. 14. Here, the left turbine normal to the flow is displayed in grey and the right turbine normal to the flow is displayed in gold. It should be noted that both turbines are rotating in clockwise direction with respect to the freestream. The red line represents the system average for the two turbines and the blue line represents the results of the disk testing. As was seen in the disk testing, the peak increase in drag of 15% was seen at an X/D of 0.2. However, the wind lenses displayed a peak increase of 16% at an X/D of 0.1. This result implies that the turbines can be placed more compactly in the grid to increase the power which reduces the footprint of the grid system. It can also be seen in Fig. 14 that the right turbine outperforms the left turbine. This bias could be attributed to the swirl component of the flow created by the rotation of the turbine. It is hypothesized that the clockwise swirl component of the left turbine induces additional velocity on the right turbine causing the increment in Cp.

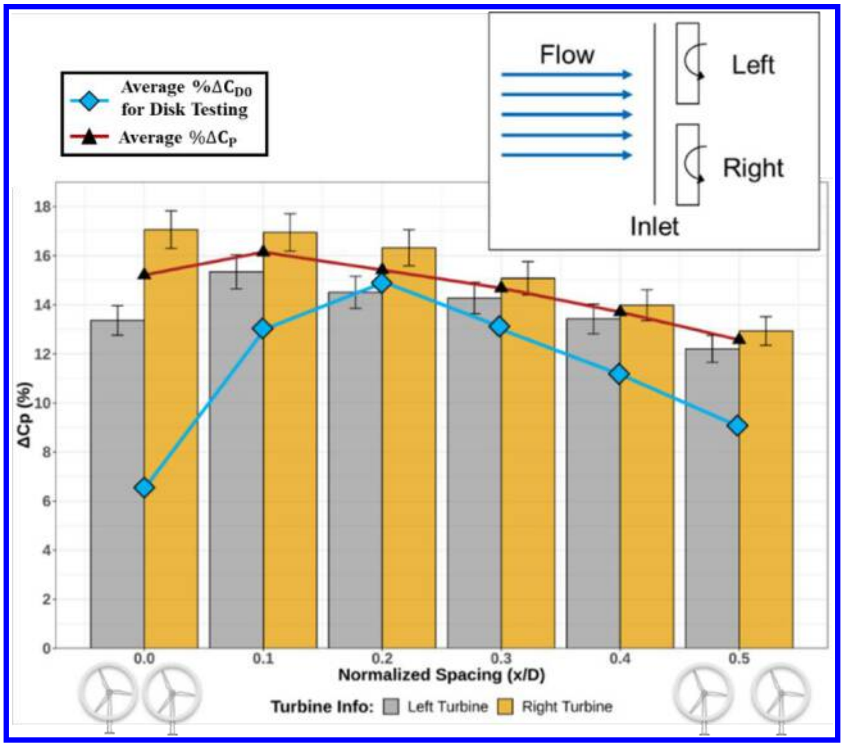


Fig. 14 Percent Change in Coefficient of Power for 1D Testing of WLTs

C) 3-D Testing

a) 3D Disk Testing (Upstream-Downstream Arrangement of the Disks)

The results of the 3D disk study are displayed in Fig. 15. The Fig 15a displays the percent change in coefficient of drag for the upstream disk at different locations of the downstream disk and Fig 15b displays the percent change in coefficient of drag on the downstream disk at different locations of the downstream disk. It should be noted once again that throughout the experiment, the upstream disk was kept in a fixed position and the downstream disk position was changed. In both images, the bright yellow region indicates an area of favorable change in drag. For the upstream disk, there is a positive increase on the order of 30%. Conversely, for the downstream disk the yellow region represents areas of little change in the coefficient of drag (~0%). The upstream disk experiences an increase in drag when the downstream disk is within 2 X/D or less. The greatest increase is seen when the downstream disk is at +/- 0.5 Y/D and 1.0 X/D. When examining the downstream disk, there is a clear area where the upstream wake of disk 1 affects the downstream disk. The upstream wake appears to extend nearly 7 diameters downstream and 1 diameter in the cross-flow direction. Therefore, in theory, the downstream turbine can be placed outside of the parabolic wake region of the upstream disk to operate nominally. The turbulent intensities in these regions however will be greater than the freestream. One interesting behavior of the downstream disk is the suction effect that it is seeing at 0.0 Y/D and 1.0 X/D. In this area, the percent change is greater than -100% which means that the downstream disk is essentially generating thrust. It is believed that this can be explained the strong area of low pressure and recirculation region immediately behind the upstream disk and the flow reestablishing itself behind the downstream disk.

b) 3D WLT Testing (Upstream-Downstream Arrangement of the WLTs)

The results of the 3D testing for the WLTs turbines are displayed in Fig. 16. The percent change in coefficient of power for the upstream and downstream turbine is shown in the left and right image, respectively. It should be noted that a maximum downstream location of only 2 Y/D could be achieved in the WLT testing whereas a maximum downstream location of 8 Y/D could be achieved during the disk testing. Similar to the results of the disk testing, the upstream turbine is affected by a downstream turbine that is in close proximity. When the downstream turbine is directly behind the upstream turbine (X/D: 0.5, Y/D: 0), the peak power decrement is seen at 18%. Additionally, at this location, the downstream turbine sees a large decrement in power at 90%. However, as the downstream turbine moves away from the upstream turbine, in both X/D and Y/D, both turbines start to recover their power. Unlike the results from the disk testing, the percent change in coefficient of power is not symmetric about the y-axis for the 3D WLT testing. This bias effect is believed to be due to the rotation of the blades and the swirl component of the wake. The upstream turbine is affected more if the downstream turbine is in the retreating blade side when compared to the advancing blade side. As shown in the single turbine testing, the rotation speed of the turbines is on the order of 1000 RPM. Therefore, the swirl of the wake may be biasing the downstream flow causing the nonsymmetrical behavior. The implications of this are that compact, 3D grid arrays must consider the rotation of direction of the blades as this will impact the overall performance of the structure. Thus, further work need be done to understand the effect of blade rotation on a large-scale structure and the best setup for many WLTs in close proximity.

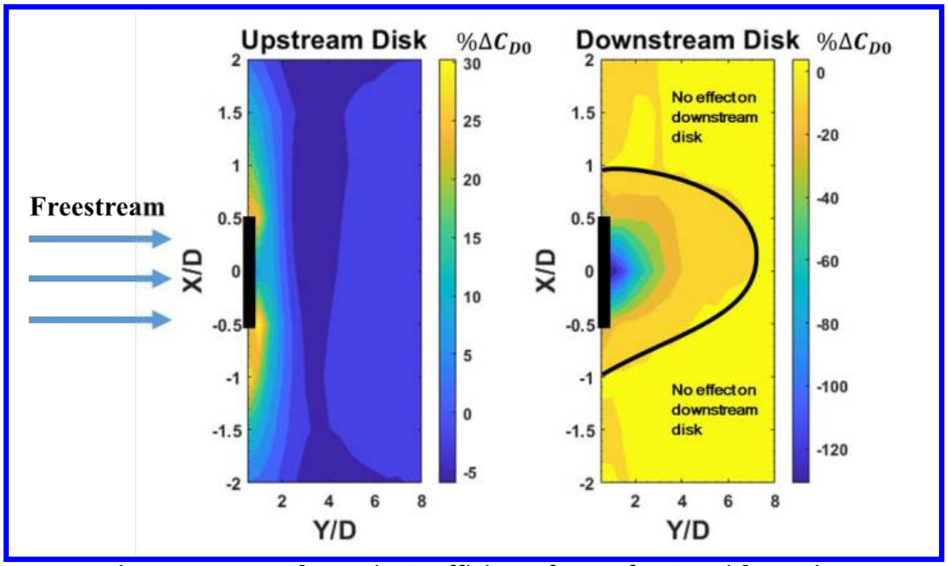


Fig. 15 Percent Change in Coefficient of Drag for 3D Disk Testing

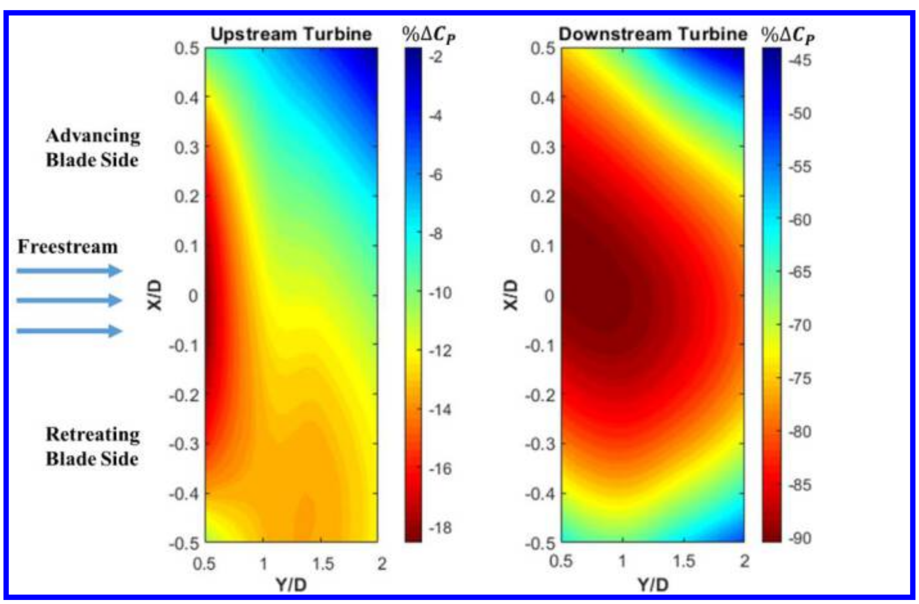


Fig. 16 Percent Change in Coefficient of Power for 3D WLT Testing

D. Field Testing a Grid Array of 12 WLTs

The results from the field test of 12 WLTs turbines is shown in Fig. 17. The y-axis displays the electrical power output of the field grid array compared with the results from wind tunnel testing and a standalone WLT in the field placed far from the grid (used as baseline/control). The wind tunnel results shown in Fig 17 are determined by multiplying independent turbine performance tested in the wind tunnel (from Fig. 12b) by a factor of 12. A 14% increase in coefficient of power was also added to account for close proximity of the WLTs in a theoretical grid. In addition to the power results, the wind distribution during testing is displayed. The power and wind data was sampled at 2 Hz over ~2 days for a total of 339,046 data points. Over the course of testing, the average wind speed was 2.4 m/s. The wind data is positively skewed which shows that most of the data was below the mean speed of 2.4 m/s but occasional gusts shifted the mean to the right, skewing the distribution. The median of the distribution was found to be 2.2 m/s. The max wind speed was 12.1 m/s.

The wind tunnel WLT results represents an ideal case with no atmospheric turbulence or boundary layer effects. Therefore, it was expected that the field testing would display a decrease in power compared to the wind tunnel. However, compared to the control turbine, the grid array performed much better at speeds greater than 6 m/s displaying that a grid structure is far more beneficial than individual turbines. At a speed of 12 m/s, the grid array displayed an increase in power production by a factor of 2.85 when compared to the control turbine. Since, the control turbine represents the performance of 12 individual WLTs without the privilege of aerodynamic interactions, the increase in power generated is purely a function of aerodynamic interaction between the wind lens turbines in the grid structure. This evinces that the grid displays superior power production to the independent turbine configuration, particularly at higher speeds. However, an interesting result of the field study displayed that all three turbines configurations performed similarly at the median wind speed of 2.2 m/s. At this point, the turbines are just beyond the point of their respective startup speeds and there is very little difference in the power produced. Another benefit of the grid structure over individual turbines is that the grid allows the turbines to naturally reach higher altitudes by stacking the turbines in the grid. This makes the turbines more effective as they are able further escape the atmospheric boundary layer.

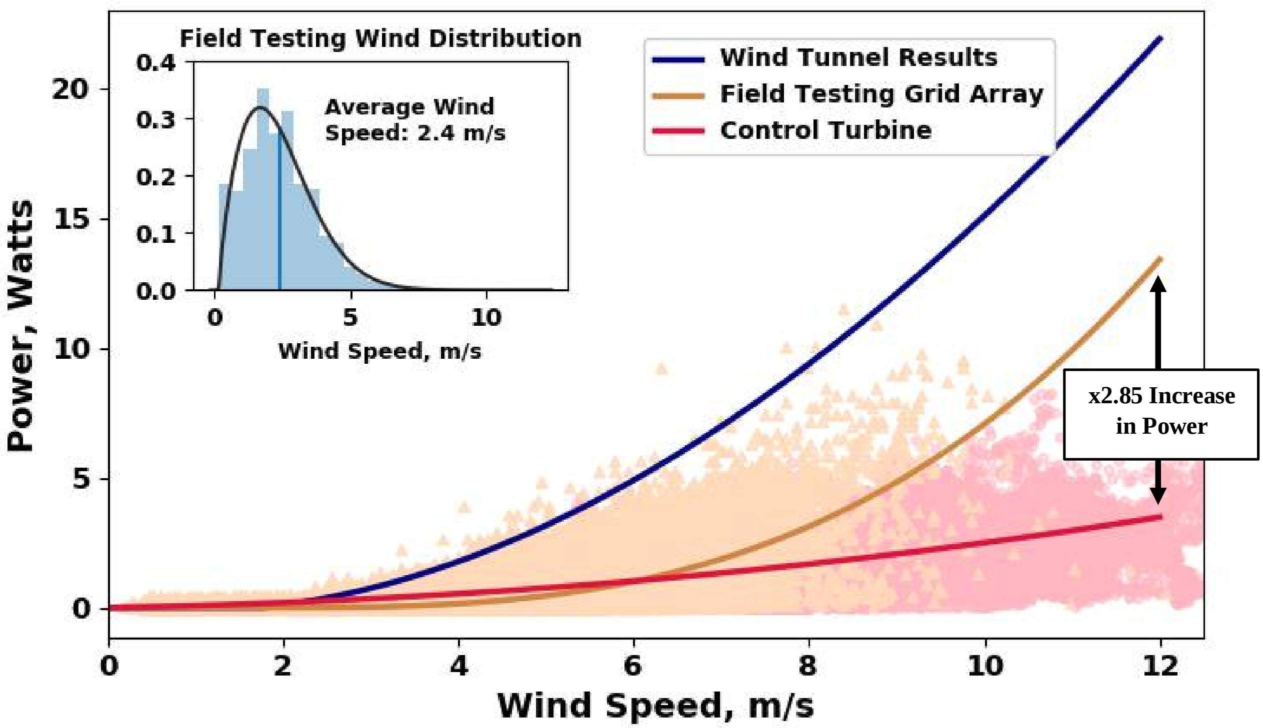


Fig. 17 Field Testing Results Comparing Wind Tunnel Data, Field Grid Array, and Field Control Turbine

VI. Conclusions

The performance and proximity investigations on 1-foot WLTs were conducted in all three axes. The WLT testing campaign was paralleled with a similar circular disk testing campaign to provide insights on the lens-lens aerodynamic interactions. The performance of the WLTs grid system was also investigated through field test. It was determined that:

- The low Reynolds number optimized WLT blade resulted in a C_P of 10.6% obtained through wind tunnel testing.
- 1D disk and 1D WLT testing displayed reasonable agreement with each other and with previous literature. One major implication of this agreement is that the disk testing can be used to model the interactions of WLTs at any scale.
- 1D WLT testing displayed an increase in the coefficient of power of 16% at an X/D value of 0.1 which resulted in an increased C_P of the rotor from 10.6% to 12.3% when placed in a grid.
- The 3-D disk testing indicated a higher stagnation zone behind the disk as expected and that the wake begins to recover around 7 diameters downstream. The 3-D WLT showed similar decrease in downstream turbine power at close proximity however, there seems to be an inherent bias in the power production. The upstream turbine shows decrement in performance if the downstream turbine is in the retreating blade side when compared to the advancing blade side.
- The field testing of 2-D grid array consisting of 12 WLTs validated the effectiveness of the grid configuration for power production. Given the non-ideal conditions, such as unsteady flows, atmospheric boundary layer and other environmental factors, the field performance of the grid system shows promise as viable option for large scale power generation.

Using the arguments presented by Jamieson [8], multi-rotor configurations drive down cost and improve quality through the standardization of parts. However, this argument can be further developed by creating small turbines that can benefit from mass manufacturing methods such as injection molding. These small turbines, such as the ones presented in this work, can substantially lower prices for energy production by taking advantage of mass manufacturing methods (injection molding), simple and available materials (plastic can be used instead of advanced composites), improved quality (standardization of parts), lower transportation costs (hundreds of units can be shipped on a box truck), and lower installation costs (requires less skilled labor), ultimately leading to a more sustainable future and achieving the goal of cost effective wind energy.

VII. Future Work

Future work for further development of small-scale wind lens turbines includes studying the WLTs in the shear layer, as well as studying the vortex-vortex interaction for lensed turbines in proximity. Understanding the effect of the shear layer allows for better development of grid structures for determining the height turbines need to be located off the ground. Particle image velocimetry (PIV) will also be performed to understand the vortex-vortex interaction and its effect on the increase in power production.

Acknowledgments

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